

```
sandysanjay817@cs-709519171448-default:~$ uname
Linux
sandysanjay817@cs-709519171448-default:~$ uptime
 08:48:52 up 15 min,  0 user,   load average: 0.03, 0.08, 0.08
sandysanjay817@cs-709519171448-default:~$ hostname
cs-709519171448-default
sandysanjay817@cs-709519171448-default:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
sandysanjay817@cs-709519171448-default:~$ free
              total        used        free      shared  buff/cache   available
Mem:           8138096       1416060       5534480          1144       1434364       6722036
Swap:              0              0              0
```

```
top - 08:48:08 up 14 min,  0 user,   load average: 0.07, 0.10, 0.08
Tasks: 29 total,  1 running, 28 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  4.0 sy,  0.0 ni, 88.0 id,  8.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7947.4 total,  5407.2 free, 1380.6 used, 1400.5 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 6566.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	4324	3436	3156	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3812	2452	S	0.0	0.0	0:00.02	rsyslogd
25	root	20	0	38768	27996	10896	S	0.0	0.3	0:00.59	python
26	root	20	0	6196	2616	2380	S	0.0	0.0	0:00.00	logger
67	root	10	-10	12024	3988	2872	S	0.0	0.0	0:00.00	sshd
333	root	20	0	2021836	84684	61824	S	0.0	1.0	0:00.41	dockerd
430	root	20	0	1234012	12372	8292	S	0.0	0.2	0:00.26	editor-proxy
431	root	20	0	16328	6416	5492	S	0.0	0.1	0:00.01	sudo

```
top - 08:50:45 up 17 min,  0 user,   load average: 0.00, 0.05, 0.07
Tasks: 30 total,  1 running, 28 sleeping,  1 stopped,  0 zombie
%Cpu(s):  0.7 us,  0.5 sy,  0.0 ni, 98.7 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem : 7947.4 total,  5404.8 free, 1382.8 used, 1400.8 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 6564.5 avail Mem
```

```
0[|] 2.0% Tasks: 32, 64 thr, 0 kthr; 1 running
1[|] 0.7% Load average: 0.00 0.04 0.06
Mem[|||||||] 1.12G/7.76G Uptime: 00:17:58
Swp[ 0K/0K]
```

Main	I/O	PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU	%MEM	TIME+	Command
		713	sandysanja	10	-10	7280	2768	2004	S	0.7	0.0	0:00.07	tmux new-session -A -D -n cloudshel
		1	root	20	0	4324	3436	3156	S	0.0	0.0	0:00.07	/bin/bash /google/scripts/onrun.sh
		9	syslog	20	0	216M	3812	2452	S	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
		10	syslog	20	0	216M	3812	2452	S	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
		11	syslog	20	0	216M	3812	2452	S	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
		12	syslog	20	0	216M	3812	2452	S	0.0	0.0	0:00.00	/usr/sbin/rsyslogd
		25	root	20	0	38768	27996	10896	S	0.0	0.3	0:00.64	/usr/bin/python /usr/bin/supervisor
		26	root	20	0	6196	2616	2380	S	0.0	0.0	0:00.00	logger -t supervisor
		67	root	10	-10	12024	3988	2872	S	0.0	0.0	0:00.00	sshd: /usr/sbin/sshd -p 22 -o Autho
		333	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.20	/usr/bin/dockerd -p /var/run/docker
		407	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.04	/usr/bin/dockerd -p /var/run/docker
		408	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.04	/usr/bin/dockerd -p /var/run/docker
		409	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.01	/usr/bin/dockerd -p /var/run/docker
		410	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker
		430	root	20	0	1205M	12372	8292	S	0.0	0.2	0:00.04	/google/devshell/editor/proxy/edito
		431	root	20	0	16328	6416	5492	S	0.0	0.1	0:00.01	sudo /google/devshell/tmux-agent
		440	root	20	0	1205M	12372	8292	S	0.0	0.2	0:00.04	/google/devshell/editor/proxy/edito
		441	root	20	0	1205M	12372	8292	S	0.0	0.2	0:00.02	/google/devshell/editor/proxy/edito
		442	root	20	0	1205M	12372	8292	S	0.0	0.2	0:00.00	/google/devshell/editor/proxy/edito
		443	root	20	0	1197M	2704	1980	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
		453	root	20	0	1197M	2704	1980	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
		454	root	20	0	1197M	2704	1980	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
		455	root	20	0	1197M	2704	1980	S	0.0	0.0	0:00.00	/google/devshell/tmux-agent
		470	root	20	0	1205M	12372	8292	S	0.0	0.2	0:00.05	/google/devshell/editor/proxy/edito
		471	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.00	/usr/bin/dockerd -p /var/run/docker
		472	root	20	0	1974M	84716	61824	S	0.0	1.0	0:00.01	/usr/bin/dockerd -p /var/run/docker
		474	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.06	containerd --config /var/run/docker
		476	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.35	containerd --config /var/run/docker
		477	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.25	containerd --config /var/run/docker
		478	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.29	containerd --config /var/run/docker
		479	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.00	containerd --config /var/run/docker
		480	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.00	containerd --config /var/run/docker
		494	root	20	0	1678M	45684	32256	S	0.0	0.6	0:00.26	containerd --config /var/run/docker

```
[2]+  Stopped                  top
sandysanjay817@cs-709519171448-default:~$ htop

[3]+  Stopped                  htop
sandysanjay817@cs-709519171448-default:~$ htop

[4]+  Stopped                  htop
sandysanjay817@cs-709519171448-default:~$ lsb_release
No LSB modules are available.
sandysanjay817@cs-709519171448-default:~$ arch
x86_64
sandysanjay817@cs-709519171448-default:~$ uname
Linux
sandysanjay817@cs-709519171448-default:~$ ls
file.txt  README-cloudshell.txt  sandy  sanjay  sid.py
sandysanjay817@cs-709519171448-default:~$ cd sandy
sandysanjay817@cs-709519171448-default:~/sandy$ pwd
/home/sandysanjay817/sandy
sandysanjay817@cs-709519171448-default:~/sandy$ cd ~
sandysanjay817@cs-709519171448-default:~$ mkdir sanjay
mkdir: cannot create directory 'sanjay': File exists
sandysanjay817@cs-709519171448-default:~$ rm sanjay
sandysanjay817@cs-709519171448-default:~$ ls
file.txt  README-cloudshell.txt  sandy  sid.py
sandysanjay817@cs-709519171448-default:~$ mkdir sanjay
sandysanjay817@cs-709519171448-default:~$ ls
file.txt  README-cloudshell.txt  sandy  sanjay  sid.py
sandysanjay817@cs-709519171448-default:~$ rmdir sanjay
sandysanjay817@cs-709519171448-default:~$ ls
file.txt  README-cloudshell.txt  sandy  sid.py
sandysanjay817@cs-709519171448-default:~$ touch sample
sandysanjay817@cs-709519171448-default:~$ ls
file.txt  README-cloudshell.txt  sample  sandy  sid.py
sandysanjay817@cs-709519171448-default:~$ mv sample sandy/
sandysanjay817@cs-709519171448-default:~$ cd sandy
sandysanjay817@cs-709519171448-default:~/sandy$ ls
sample  sid2.py
sandysanjay817@cs-709519171448-default:~/sandy$ cd ~
sandysanjay817@cs-709519171448-default:~$ touch sample
sandysanjay817@cs-709519171448-default:~$ cat sample
sandysanjay817@cs-709519171448-default:~$ cat sid.py
```

```
sandysanjay817@cs-709519171448-default:~$ cat sid.py
#first line
print("h")
#last line

#last part
ok done

sandysanjay817@cs-709519171448-default:~$ cp sid.py sample
sandysanjay817@cs-709519171448-default:~$ cat sample
#first line
print("h")
#last line

#last part
ok done
```

```
sandysanjay817@cs-709519171448-default:~$ ls
file.txt  README-cloudshell.txt  sample  sandy  sid.py
sandysanjay817@cs-709519171448-default:~$ ln sample sid.py
ln: failed to create hard link 'sid.py': File exists
sandysanjay817@cs-709519171448-default:~$ ln sid.py file
sandysanjay817@cs-709519171448-default:~$ ls
file  file.txt  README-cloudshell.txt  sample  sandy  sid.py
sandysanjay817@cs-709519171448-default:~$ ls -l
file
file.txt
README-cloudshell.txt
sample
sandy
sid.py
sandysanjay817@cs-709519171448-default:~$ cat sid.py
#first line
print("h")
#last line

#last part
ok done

sandysanjay817@cs-709519171448-default:~$ more file sample
:::::::::::::
file
:::::::::::::
#first line
print("h")
#last line

#last part
ok done

:::::::::::::
sample
:::::::::::::
#first line
print("h")
#last line
```

```
sandysanjay817@cs-709519171448-default:~$ less file
```

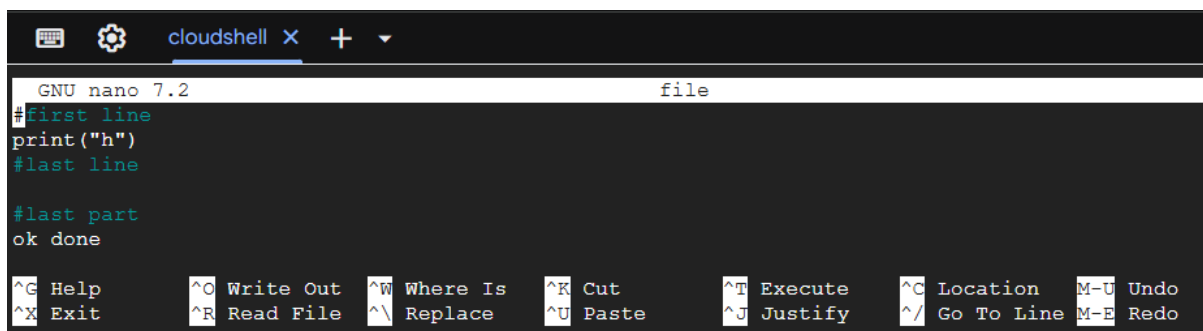
```
[5]+  Stopped                  less file
sandysanjay817@cs-709519171448-default:~$ tail file
#first line
print("h")
#last line

#last part
ok done
```

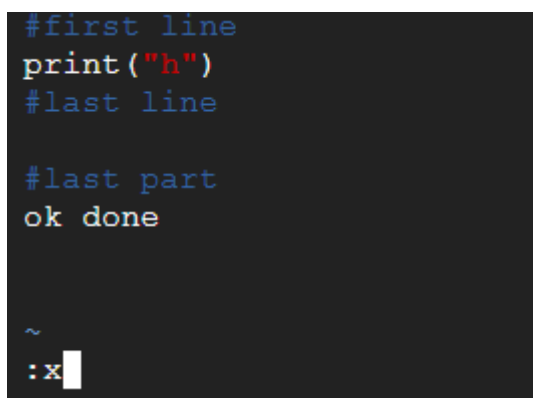
```
sandysanjay817@cs-709519171448-default:~$ head file
#first line
print("h")
#last line

#last part
ok done
```

```
sandysanjay817@cs-709519171448-default:~$ nano file
```



The screenshot shows a cloudshell terminal window with a dark background. At the top, there's a header bar with a keyboard icon, a gear icon, and the text 'cloudshell' followed by a close button 'X' and a plus sign with a dropdown arrow. Below this, the terminal shows the 'GNU nano 7.2' prompt and the filename 'file'. The content of the file is displayed in a light blue monospace font: '#first line', 'print("h")', '#last line', a blank line, '#last part', and 'ok done'. At the bottom of the terminal, there's a horizontal menu of keyboard shortcuts in a light blue font: '^G Help', '^O Write Out', '^W Where Is', '^K Cut', '^T Execute', '^C Location', 'M-U Undo', '^X Exit', '^R Read File', '^_ Replace', '^U Paste', '^J Justify', '^/ Go To Line', and 'M-E Redo'.



This is a close-up screenshot of the nano text editor's content area. It shows the same text as the previous screenshot: '#first line', 'print("h")', '#last line', a blank line, '#last part', and 'ok done'. The text is in a light blue monospace font on a dark background. At the bottom left, there's a tilde '~' followed by a prompt ':x' and a white cursor block.

```
sandysanjay817@cs-709519171448-default:~$ ln sid.py file
sandysanjay817@cs-709519171448-default:~$ gedit
-bash: gedit: command not found
sandysanjay817@cs-709519171448-default:~$ awk '{print}' file
#first line
print("h")
#last line

#last part
ok done

sandysanjay817@cs-709519171448-default:~$ awk '{print $2}' file
line

line

part
done

sandysanjay817@cs-709519171448-default:~$ sed file
sed: -e expression #1, char 1: unknown command: `f'
sandysanjay817@cs-709519171448-default:~$ sed '2d' file.txt
sandysanjay817@cs-709519171448-default:~$ cat file
#first line
print("h")
#last line

#last part
ok done

sandysanjay817@cs-709519171448-default:~$ sed '2d' file
#first line
#last line

#last part
ok done

sandysanjay817@cs-709519171448-default:~$
```

```
sandysanjay817@cs-709519171448-default:~$ chmod 777 file
sandysanjay817@cs-709519171448-default:~$ chmod -w file
chmod: file: new permissions are r-xr-xrwx, not r-xr-xr-x
sandysanjay817@cs-709519171448-default:~$ chmod -r
chmod: missing operand
Try 'chmod --help' for more information.
sandysanjay817@cs-709519171448-default:~$ chmod -r file
sandysanjay817@cs-709519171448-default:~$
```

```

sandysanjay817@cs-709519171448-default:~$ chmod 777 file
sandysanjay817@cs-709519171448-default:~$ chmod -w file
chmod: file: new permissions are r-xr-xrwx, not r-xr-xr-x
sandysanjay817@cs-709519171448-default:~$ chmod -r
chmod: missing operand
Try 'chmod --help' for more information.
sandysanjay817@cs-709519171448-default:~$ chmod -r file
sandysanjay817@cs-709519171448-default:~$ chown sandy file
chown: invalid user: 'sandy'
sandysanjay817@cs-709519171448-default:~$ whoami
sandysanjay817
sandysanjay817@cs-709519171448-default:~$ chown sandy file^C
sandysanjay817@cs-709519171448-default:~$ chown sandysanjay817 file
sandysanjay817@cs-709519171448-default:~$ ls -l file
---x--x-wx 2 sandysanjay817 sandysanjay817 56 May 14 07:24 file
sandysanjay817@cs-709519171448-default:~$ █

```

```

sandysanjay817@cs-709519171448-default:~$ chmod 777 file
sandysanjay817@cs-709519171448-default:~$ chmod -w file
chmod: file: new permissions are r-xr-xrwx, not r-xr-xr-x
sandysanjay817@cs-709519171448-default:~$ chmod -r
chmod: missing operand
Try 'chmod --help' for more information.
sandysanjay817@cs-709519171448-default:~$ chmod -r file
sandysanjay817@cs-709519171448-default:~$ chown sandy file
chown: invalid user: 'sandy'
sandysanjay817@cs-709519171448-default:~$ whoami
sandysanjay817
sandysanjay817@cs-709519171448-default:~$ chown sandy file^C
sandysanjay817@cs-709519171448-default:~$ chown sandysanjay817 file
sandysanjay817@cs-709519171448-default:~$ ls -l file
---x--x-wx 2 sandysanjay817 sandysanjay817 56 May 14 07:24 file
sandysanjay817@cs-709519171448-default:~$ sudo adduser sandy
info: Adding user `sandy' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `sandy' (1001) ...
info: Adding new user `sandy' (1001) with group `sandy (1001)' ...
info: Creating home directory `/home/sandy' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sandy
Enter the new value, or press ENTER for the default
    Full Name []: sandy
    Room Number []: 2
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `sandy' to supplemental / extra groups `users' ...
info: Adding user `sandy' to group `users' ...
sandysanjay817@cs-709519171448-default:~$ chown sandy file
chown: changing ownership of 'file': Operation not permitted
sandysanjay817@cs-709519171448-default:~$ ls -l file
---x--x-wx 2 sandysanjay817 sandysanjay817 56 May 14 07:24 file
sandysanjay817@cs-709519171448-default:~$ sudo chown sandy file
sandysanjay817@cs-709519171448-default:~$ ls -l file
---x--x-wx 2 sandy sandysanjay817 56 May 14 07:24 file

```



```
sandysanjay817@cs-709519171448-default:~$ ls
file file.txt README-cloudshell.txt sample sandy sid.py
sandysanjay817@cs-709519171448-default:~$ sudo groupadd developers
sandysanjay817@cs-709519171448-default:~$ groups
sandysanjay817 adm sudo docker
sandysanjay817@cs-709519171448-default:~$ sudo systemctl unmask service_name
sandysanjay817@cs-709519171448-default:~$ systemctl list-unit-files --state=masked
```

UNIT	FILE	STATE	PRESET
cryptdisks-early.service		masked	enabled
cryptdisks.service		masked	enabled
hwclock.service		masked	enabled
sudo.service		masked	enabled
x11-common.service		masked	enabled

```
5 unit files listed.
sandysanjay817@cs-709519171448-default:~$ stat file
  File: file
  Size: 56          Blocks: 8          IO Block: 4096   regular file
Device: 8,17      Inode: 131275       Links: 2
Access: (0113/---x--x-wx)  Uid: ( 1001/ UNKNOWN)   Gid: ( 1000/sandysanjay817)
Access: 2026-05-14 07:24:34.271170097 +0000
Modify: 2026-05-14 07:24:34.271170097 +0000
Change: 2026-05-14 10:42:51.377940754 +0000
 Birth: 2026-05-14 07:24:34.271170097 +0000
sandysanjay817@cs-709519171448-default:~$ getfacl file
-bash: getfacl: command not found
sandysanjay817@cs-709519171448-default:~$ getfacl sid.py
-bash: getfacl: command not found
sandysanjay817@cs-709519171448-default:~$ sudo apt update
sudo apt install acl
```

```
sandysanjay817@cs-709519171448-default:~$ which getfacl
/usr/bin/getfacl
sandysanjay817@cs-709519171448-default:~$ getfacl sid.py
# file: sid.py
# owner: 1001
# group: sandysanjay817
user::--x
group::--x
other::--wx
```

```
sandysanjay817@cs-709519171448-default:~$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | amt | fou | help | ila | ioam | l2tp |
                  link | macsec | maddress | monitor | mptcp | mroute | mrule |
                  neighbor | neighbour | netconf | netns | nexthop | ntable |
                  ntbl | route | rule | sr | tap | tcpmetrics |
                  token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                   -h[uman-readable] | -iec | -j[son] | -p[retty] |
                   -f[amily] { inet | inet6 | mpls | bridge | link } |
                   -4 | -6 | -M | -B | -0 |
                   -l[oops] { maximum-addr-flush-attempts } | -br[ief] |
                   -af[family] | -t[imeout] | -t[oken] | -b[atch] | -f[ile] | -f[ile] | -f[ile] }
```

```
sandysanjay817@cs-709519171448-default:~$ ping google.com
PING google.com (142.251.10.100) 56(84) bytes of data:
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=1 ttl=114 time=0.729 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=2 ttl=114 time=0.499 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=3 ttl=114 time=0.520 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=4 ttl=114 time=0.423 ms
^Z
[1]+  Stopped                  ping google.com
```

```
sandysanjay817@cs-709519171448-default:~$ netstat
Active Internet connections (w/o servers)
```

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	localhost:54180	localhost:970	ESTABLISHED
tcp	0	0	localhost:35260	localhost:ssh	ESTABLISHED
tcp	0	0	localhost:ssh	localhost:35264	ESTABLISHED
tcp	0	0	localhost:ssh	localhost:35260	ESTABLISHED
tcp	0	0	localhost:55422	localhost:971	ESTABLISHED

```
sandysanjay817@cs-709519171448-default:~$ lsattr file
lsattr: Permission denied While reading flags on file
sandysanjay817@cs-709519171448-default:~$ sudo lsattr file
-----e----- file
sandysanjay817@cs-709519171448-default:~$ sudo chattr -i file
sandysanjay817@cs-709519171448-default:~$ sudo lsattr file
-----e----- file
sandysanjay817@cs-709519171448-default:~$ sudo chattr +ti file.txt
sandysanjay817@cs-709519171448-default:~$ sudo lsattr file
-----e----- file
sandysanjay817@cs-709519171448-default:~$ sudo chattr +a file
sandysanjay817@cs-709519171448-default:~$ touch file
touch: cannot touch 'file': Permission denied
sandysanjay817@cs-709519171448-default:~$ cat file
cat: file: Permission denied
sandysanjay817@cs-709519171448-default:~$ sudo cat file
#first line
print("h")
#last line

#last part
ok done

sandysanjay817@cs-709519171448-default:~$ sudo usermod -g developers sandy
usermod: user 'sandy' does not exist
sandysanjay817@cs-709519171448-default:~$ adduser sandy
fatal: Only root may add a user or group to the system.
sandysanjay817@cs-709519171448-default:~$ sudo adduser sandy
info: Adding user `sandy' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `sandy' (1002) ...
info: Adding new user `sandy' (1002) with group `sandy (1002)' ...
warn: The home directory `/home/sandy' already exists. Not touching this directory.
warn: Warning: The home directory `/home/sandy' does not belong to the user you are currently creating.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sandy
Enter the new value, or press ENTER for the default
    Full Name []: sandy
    Room Number []: 11
```

```
Enter the new value, or press ENTER for the default
    Full Name []: sandy
    Room Number []: 11
    Work Phone []: 12
    Home Phone []: 12
    Other []: 12
Is the information correct? [Y/n] Y
info: Adding new user `sandy' to supplemental / extra groups `users' ...
info: Adding user `sandy' to group `users' ...
sandysanjay817@cs-709519171448-default:~$ id sandy
uid=1002(sandy) gid=1002(sandy) groups=1002(sandy),100(users)
sandysanjay817@cs-709519171448-default:~$
sandysanjay817@cs-709519171448-default:~$ sudo usermod -g developers sandy
```

```
sandysanjay817@cs-709519171448-default:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether ca:a0:2d:6e:40:e5 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.4 netmask 255.255.0.0 broadcast 10.88.255.255
    ether 66:f4:af:0c:27:53 txqueuelen 1000 (Ethernet)
```

```
sandysanjay817@cs-709519171448-default:~$ ss
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
u_str ESTAB 0 0 * 21966 * 21965
u_str ESTAB 0 0 * 22757 * 22756
u_str ESTAB 0 0 * 21107 * 21108
```



```
sandysanjay817@cs-709519171448-default:~$ traceroute google.com
-bash: traceroute: command not found
sandysanjay817@cs-709519171448-default:~$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 34 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 0s (1,567 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package traceroute.
(Reading database ... 121200 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute.1.gz because associated file /usr/share/man/man1/traceroute.db.1.gz (of link group traceroute) doesn't exist
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute6.1.gz because associated file /usr/share/man/man1/traceroute6.db.1.gz (of link group traceroute6) doesn't exist
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/lft.1.gz because associated file /usr/share/man/man1/lft.db.1.gz (of link group lft) doesn't exist
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceproto.1.gz because associated file /usr/share/man/man1/traceproto.db.1.gz (of link group traceproto) doesn't exist
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man8/tcptraceroute.8.gz because associated file /usr/share/man/man8/tcptraceroute.db.8.gz (of link group tcptraceroute) doesn't exist
Processing triggers for man-db (2.12.0-4build2) ...
sandysanjay817@cs-709519171448-default:~$ traceroute -n google.com
traceroute to google.com (172.217.194.101), 30 hops max, 60 byte packets
 1  10.88.0.1  0.064 ms  0.016 ms  0.011 ms
 2  * * *
 3  172.253.68.142  0.423 ms  * *
```

```
sandysanjay817@cs-709519171448-default:~$ wget google.com
--2026-05-14 16:28:32-- http://google.com/
Resolving google.com (google.com)... 142.250.4.139, 142.250.4.101, 142.250.4.100, ...
Connecting to google.com (google.com)|142.250.4.139|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: http://www.google.com/ [following]
--2026-05-14 16:28:32-- http://www.google.com/
Resolving www.google.com (www.google.com)... 142.251.153.119, 142.251.154.119, 142.251.150.119, ...
Connecting to www.google.com (www.google.com)|142.251.153.119|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html          [ <=> ] 78.96K --.-KB/s   in 0.003s

2026-05-14 16:28:32 (22.2 MB/s) - 'index.html' saved [80857]
```

```
sandysanjay817@cs-709519171448-default:~$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-SoNlAauwwGJwHEnA0ze-Wg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
date: Thu, 14 May 2026 16:30:00 GMT
expires: Sat, 13 Jun 2026 16:30:00 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
```

```
sandysanjay817@cs-709519171448-default:~$ ps
```

PID	TTY	TIME	CMD
634	pts/2	00:00:00	bash
2118	pts/2	00:00:00	ping
2119	pts/2	00:00:00	ping
2179	pts/2	00:00:00	nslookup
2196	pts/2	00:00:00	ps

```
sandysanjay817@cs-709519171448-default:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20690
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                290     IN      A      74.125.130.139
google.com.                290     IN      A      74.125.130.113
google.com.                290     IN      A      74.125.130.102
google.com.                290     IN      A      74.125.130.101
google.com.                290     IN      A      74.125.130.138
google.com.                290     IN      A      74.125.130.100

;; Query time: 1 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 16:31:01 UTC 2026
;; MSG SIZE rcvd: 135
```

```
sandysanjay817@cs-709519171448-default:~$ nslookup google.com
Server:      169.254.169.254
Address:     169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 74.125.130.139
Name:   google.com
Address: 74.125.130.113
Name:   google.com
Address: 74.125.130.102
Name:   google.com
Address: 74.125.130.101
Name:   google.com
Address: 74.125.130.138
```

```
sandysanjay817@cs-709519171448-default:~$ ps
  PID TTY          TIME CMD
   634 pts/2    00:00:00 bash
  2118 pts/2    00:00:00 ping
  2119 pts/2    00:00:00 ping
  2179 pts/2    00:00:00 nslookup
  2201 pts/2    00:00:00 ps

sandysanjay817@cs-709519171448-default:~$ sleep 60 &
[4] 2202

sandysanjay817@cs-709519171448-default:~$ ps
  PID TTY          TIME CMD
   634 pts/2    00:00:00 bash
  2118 pts/2    00:00:00 ping
  2119 pts/2    00:00:00 ping
  2179 pts/2    00:00:00 nslookup
  2202 pts/2    00:00:00 sleep
  2203 pts/2    00:00:00 ps

sandysanjay817@cs-709519171448-default:~$ pgrep sleep
741
2202
```

```
sandysanjay817@cs-709519171448-default:~$ sleep 60 &
[1] 2341
sandysanjay817@cs-709519171448-default:~$ kill 2341
[1]+  Terminated                  sleep 60
sandysanjay817@cs-709519171448-default:~$
```

```
sandysanjay817@cs-709519171448-default:~$ sleep 300 &
sleep 300 &
[1] 2345
[2] 2346
sandysanjay817@cs-709519171448-default:~$ killall sleep
sleep(2336): Operation not permitted
[1]-  Terminated                  sleep 300
[2]+  Terminated                  sleep 300
sandysanjay817@cs-709519171448-default:~$
```

```
sandysanjay817@cs-709519171448-default:~$ sleep 100
^Z
[1]+  Stopped                      sleep 100
sandysanjay817@cs-709519171448-default:~$ bg
[1]+ sleep 100 &
```

```
sandysanjay817@cs-709519171448-default:~$ fg
sleep 100
```

```
sandysanjay817@cs-709519171448-default:~$ jobs
[1]+  Stopped                      sleep 100
```

```
sandysanjay817@cs-709519171448-default:~$ nice -n 10 sleep 100
^Z
[2]+  Stopped                      nice -n 10 sleep 100
sandysanjay817@cs-709519171448-default:~$ renice 5 2345
renice: failed to get priority for 2345 (process ID): No such process
sandysanjay817@cs-709519171448-default:~$ ps
  PID TTY          TIME CMD
  548 pts/2        00:00:00 bash
 2348 pts/2        00:00:00 sleep
 2353 pts/2        00:00:00 sleep
 2355 pts/2        00:00:00 ps
sandysanjay817@cs-709519171448-default:~$ renice 5 2355
renice: failed to get priority for 2355 (process ID): No such process
sandysanjay817@cs-709519171448-default:~$ renice 5 2353
2353 (process ID) old priority 0, new priority 5
sandysanjay817@cs-709519171448-default:~$
```